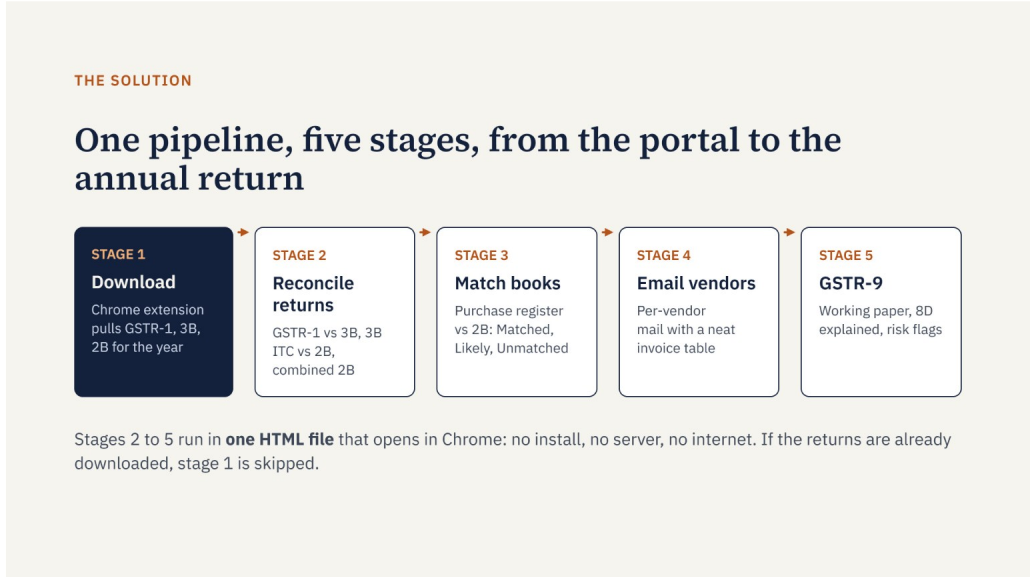


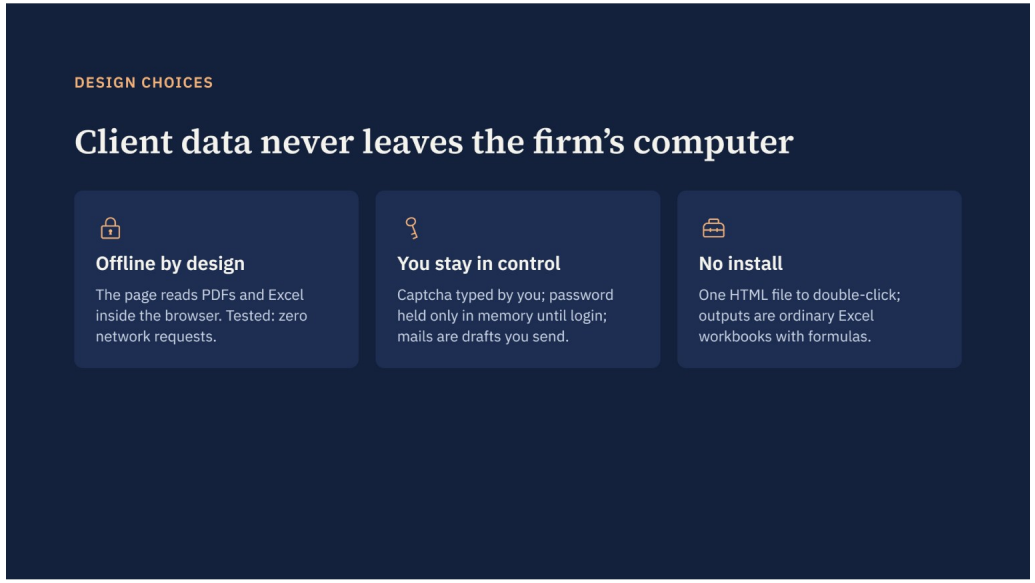
Hello, I am CA Janhavi Deshpande, from the Pune batch of the AICA Level 2 course. My capstone is an automation tool for the GST work a mid-size CA firm does every month and every year: it downloads the returns from the portal, reconciles them, matches the purchase register with GSTR-2B, prepares the vendor follow-up mails, and ends with a GSTR-9 working paper. Everything after the download runs offline, on the firm's own computer. Everything you will see today uses made-up demo data, not client data.



A mid-size firm repeats four jobs for every client. First, downloading returns: GSTR-1, 3B and 2B, thirty-six files a year per client. Second, matching the purchase register with GSTR-2B, usually with VLOOKUPs that fail the moment an invoice number is written differently. Third, writing to vendors whose invoices are missing, one mail at a time. And fourth, at year end, rebuilding GSTR-9 Tables 6 and 8 and explaining the 8D difference. None of this needs judgment until the differences are found, so it is the right place for automation.



The answer is one pipeline in five stages. Stage one is a Chrome extension that downloads the year's returns from the portal. Stages two to five live in a single HTML file: it reconciles the returns, matches the purchase register with GSTR-2B, prepares the vendor mails, and builds the GSTR-9 working paper. If someone has already downloaded the returns, they simply skip stage one and drop the files in.

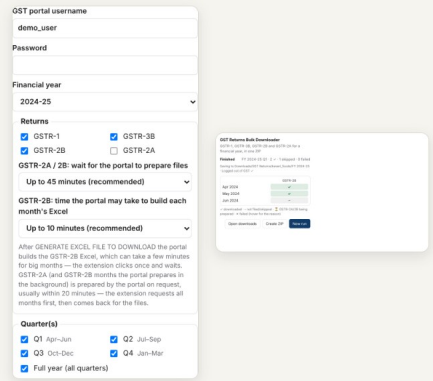


Three design choices shaped everything. First, client data never leaves the computer: the page reads the PDFs and Excel files inside the browser, and in testing it made zero network requests. Second, the professional stays in control: the captcha is typed by the user, the password is kept only in memory until login, and the vendor mails are drafts that the user reviews and sends. Third, nothing to install: one HTML file, and every output is a normal Excel workbook.

STAGE 1 · DOWNLOAD

A Chrome extension fetches the whole year in one run

- Pick the year, quarters and returns: GSTR-1, GSTR-3B, GSTR-2B
- Files already in the folder are skipped
- Waits for slow 2B builds instead of re-clicking
- Tested live: Q1 and Q2 GSTR-2B FY 2025-26 downloaded

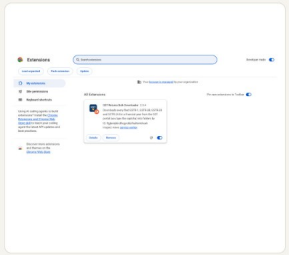


Stage one is the Chrome extension. I enter the portal username, then choose the financial year, the quarters and the returns I want: GSTR-1, 3B and 2B. I log in myself and type the captcha; the extension then walks the portal and saves each file. If a file is already in the folder it is skipped, so a second run only fetches what is missing. GSTR-2B sometimes takes a while to build, so the extension waits rather than clicking again. I tested this on the live portal and it downloaded Q1 and Q2 GSTR-2B for 2025-26 without any help.

STAGE 1 WALKTHROUGH · ONE-TIME SETUP

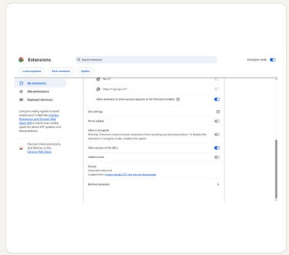
Three settings in Chrome, done once

1 Load the extension



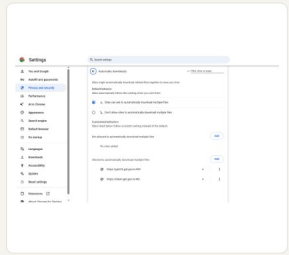
Developer mode → Load unpacked → choose the extension folder

2 Allow file URLs



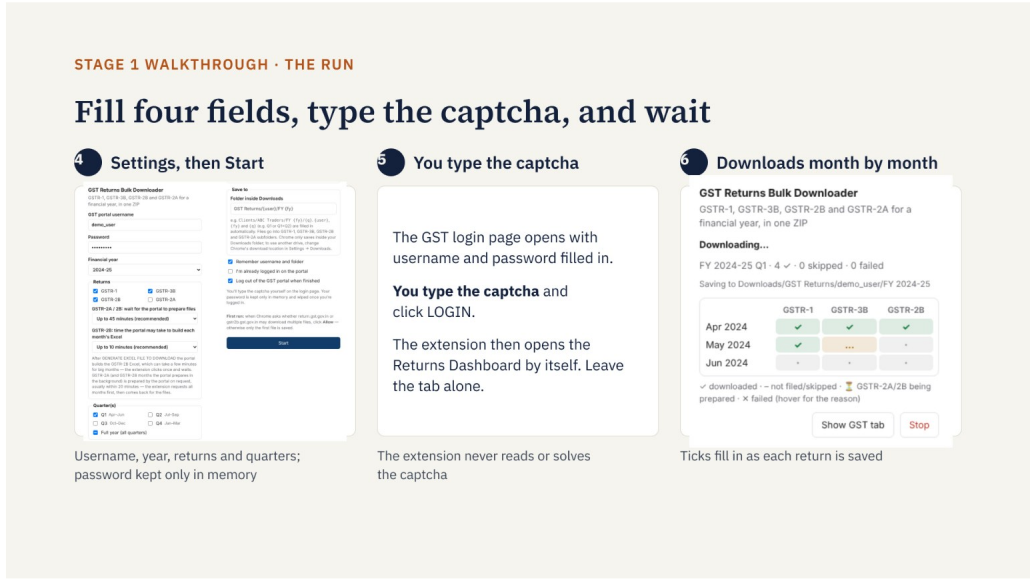
Lets the extension read the saved files to build the ZIP

3 Allow automatic downloads

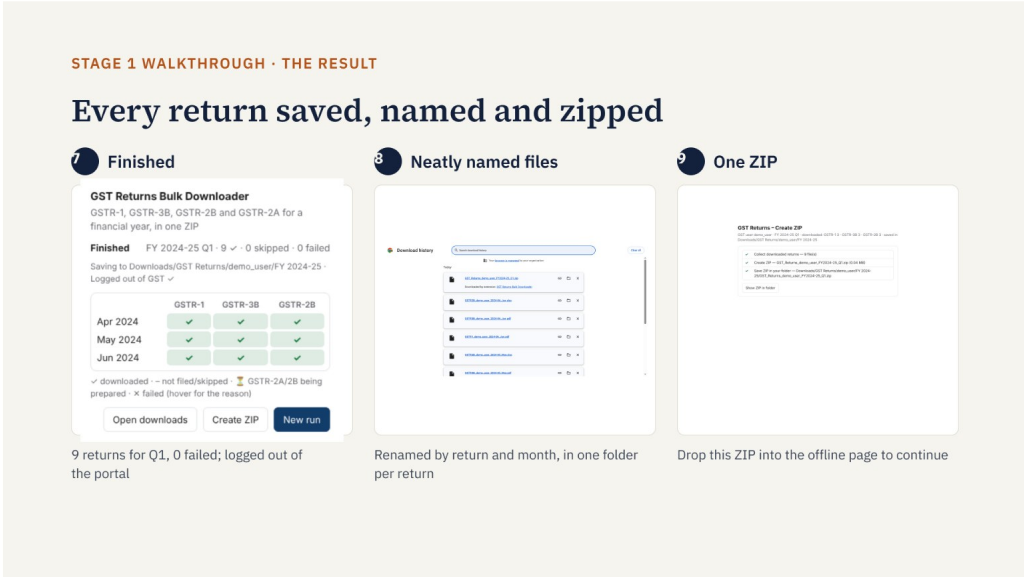


Allow return.gst.gov.in and gstr2b.gst.gov.in

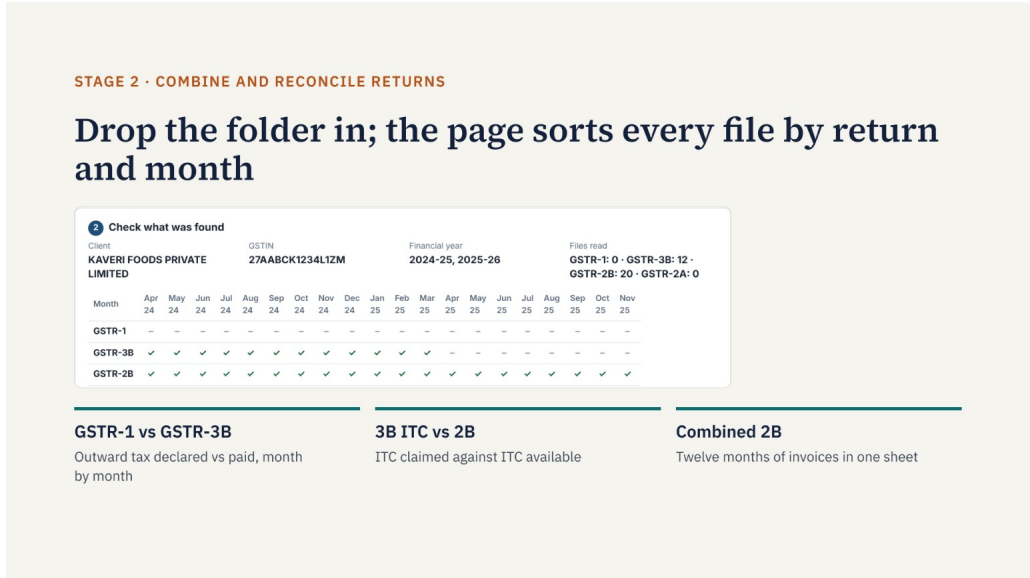
Here is the extension step by step, on made-up demo data with a test copy of the portal, so no real login appears. Setup is done once: load the extension in Chrome with Load unpacked, allow it to read file URLs so it can build the ZIP, and allow the two GST sites to download several files automatically.



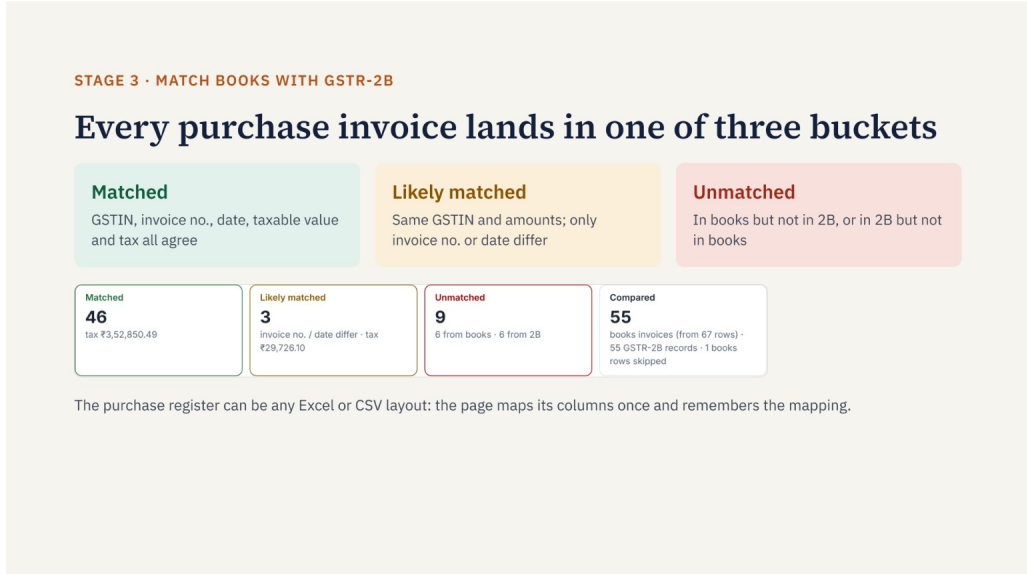
For a run, I enter the username and password, choose the year, the returns and the quarters, and click Start. The GST login page opens with the details filled in; I type the captcha myself and log in. From there the extension takes over: it opens the returns dashboard, goes month by month, and the grid fills in as each return is saved.



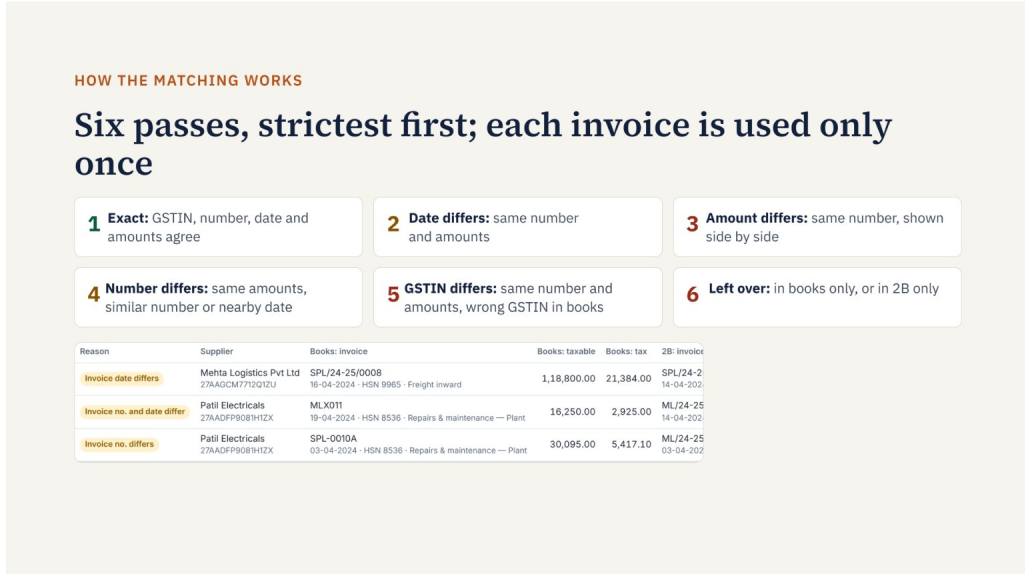
At the end, all nine returns for the quarter are downloaded with none failed, and the extension logs out of the portal. Each file is renamed by return and month and saved in its own folder; next time, files already there are skipped. Create ZIP bundles everything into one file, which goes straight into the reconciliation page, and that is stage two.



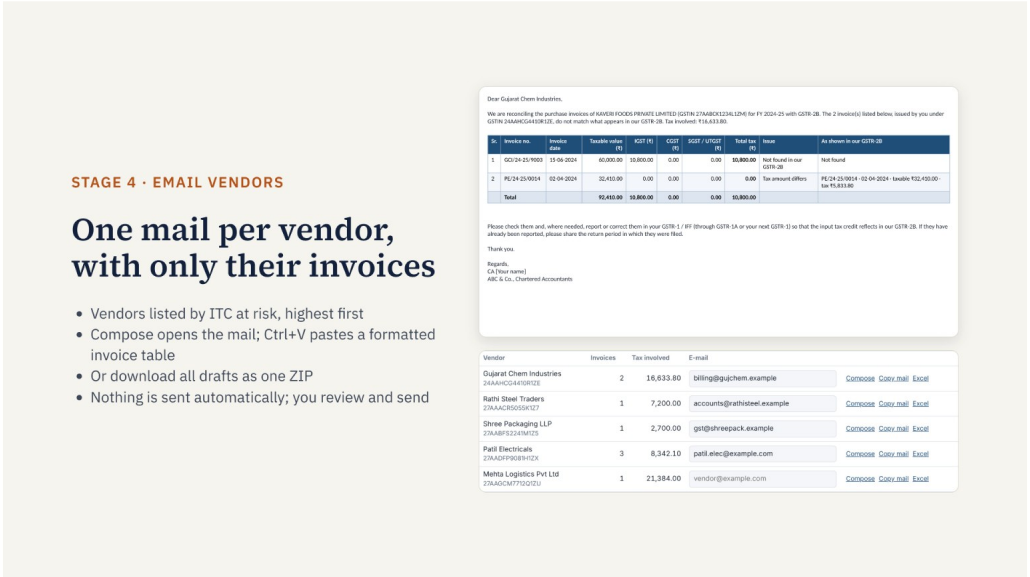
Stage two runs in the offline page. I drop the downloaded folder in, and the page recognises each file, the PDFs of GSTR-1 and 3B and the GSTR-2B Excel files, and sorts them by return and month. It then gives three reconciliations: GSTR-1 against 3B for outward tax, 3B ITC against 2B, and one combined GSTR-2B sheet with all twelve months of invoices, which is the base for the next stage.



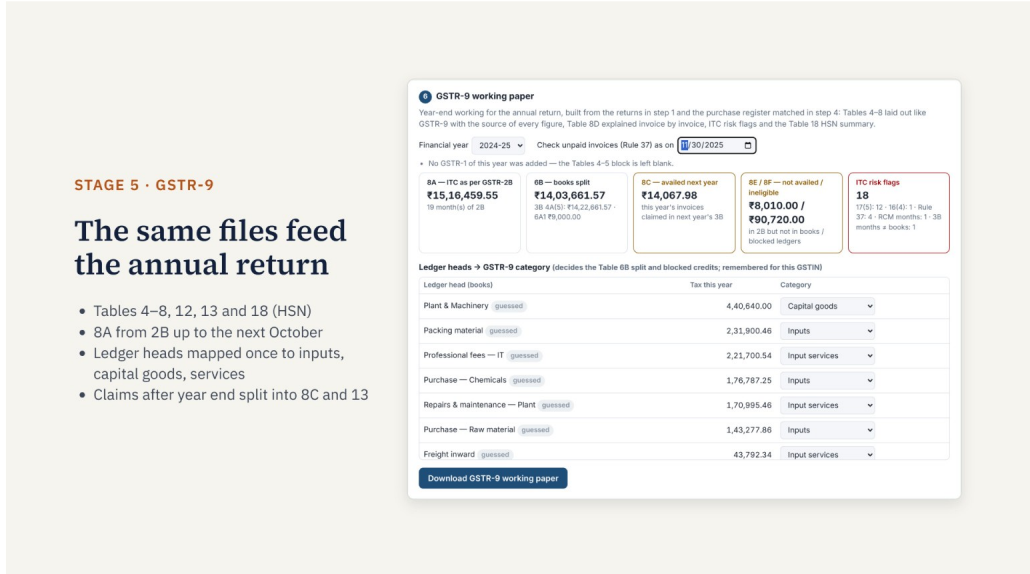
Stage three is the heart of the project: matching the purchase register with GSTR-2B. Every invoice ends up in one of three buckets. Matched, where GSTIN, invoice number, date, taxable value and tax all agree. Likely matched, where the supplier and amounts agree and only the invoice number or the date differ, which is usually a typing difference. And unmatched, which is either in our books but not in 2B, or in 2B but not in our books. The register does not need a fixed format: the page maps the columns once and remembers them for next time.



The matching runs in six passes, strictest first, and once an invoice is paired it is not used again. Pass one is the exact match. Pass two finds the same invoice number and amounts with a different date. Pass three finds the same number with different amounts and shows both sides together so the difference is obvious. Pass four catches typing differences in the invoice number, such as a missing prefix or leading zero, when the amounts agree and the date is close. Pass five catches a wrong GSTIN in our books. Whatever is left is unmatched. Every line carries the reason, as you can see here for the likely matches.



Stage four turns the results into vendor follow-up. The vendors are listed by the ITC at risk, highest first, so the team starts where the money is. For each vendor, Compose opens a mail in the default mail program, and pressing Control V pastes a neat table of only that vendor's invoices: those missing from their GSTR-1 and those where the number, date or amount differs. For a large client, all drafts can be downloaded at once as a ZIP. Nothing goes out on its own: the professional reviews every mail and sends it.

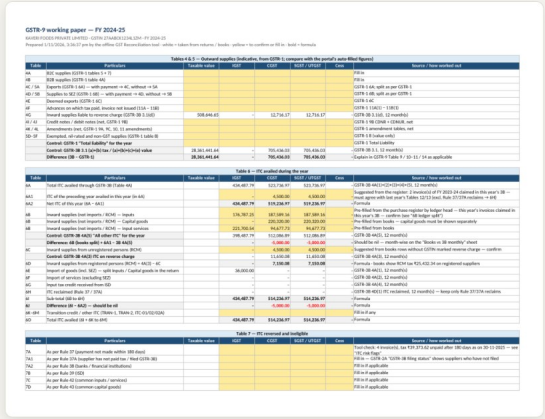


Stage five reuses the same files for the annual return. The page prepares GSTR-9 Tables 4 to 8, 12, 13 and the HSN summary in Table 18. Table 8A is built from GSTR-2B up to October of the next year, as the law allows. The purchase register's ledger heads are mapped once to inputs, capital goods and input services in Table 6, and claims made after the year end are split into 8C and Table 13 by the month they were claimed in GSTR-3B.

THE WORKING PAPER

A reviewer can trace every figure

- Yellow cells are for the CA to fill; totals are live formulas
- 8D split invoice by invoice, so “unexplained” should be nil
- Risk flags: 17(5), 16(4), Rule 37, RCM, 3B vs books
- Month-wise sheet of books vs 3B vs 2B



GSTR-9 working paper — FY 2024-25

KAVERI FOODS PRIVATE LIMITED · GSTIN 27AABCK1234L1ZM · FY 2024-25

Prepared 1/11/2026, 3:36:17 pm by the offline GST Reconciliation tool · white = taken from returns / books · yellow = to confirm or fill in · bold = formula.

Tables 4 & 5 — Outward supplies (indicative, from GSTR-1; compare with the portal's auto-filled figures)

Table	Particulars	Taxable value	IGST	CGST	SGST / UTGST	Cess	Source / how worked out
4A	B2C supplies (GSTR-1 tables 5 + 7)						Fill in
4B	B2B supplies (GSTR-1 table 4A)						Fill in
4C + 5A	Exports (GSTR-1 6A) — with payment → 4C, without → 5A						GSTR-1 6A; split as per GSTR-1
4D / 5B	Supplies to SEZ (GSTR-1 6B) — with payment → 4D, without → 5B						GSTR-1 6B; split as per GSTR-1
4E	Deemed exports (GSTR-1 6C)						GSTR-1 6C
4F	Advances on which tax paid, invoice not issued (11A – 11B)						GSTR-1 11A(1) – 11B(1)
4G	Inward supplies liable to reverse charge (GSTR-3B 3.1(d))	508,646.65	–	12,716.17	12,716.17		GSTR-3B 3.1(d); 12 months)
4I / 4J	Credit notes / debit notes (net, GSTR-1 9B)						GSTR-1 9B CDNR + CDNUR, net
4K / 4L	Amendments (net, GSTR-1 9A, 9C, 10, 11 amendments)						GSTR-1 amendment tables, net
5D–5F	Exempted, nil-rated and non-GST supplies (GSTR-1 table 8)						GSTR-1 8 (value only)
	Control: GSTR-1 "Total liability" for the year						GSTR-1 Total Liability
	Control: GSTR-3B 3.1(a)+(b)+(c)+(d)+(e)+(f)+(g) value	28,361,441.64	–	705,436.03	705,436.03		GSTR-3B 3.1; 12 months)
	Difference (3B − GSTR-1)	**28,361,441.64**	–	**705,436.03**	**705,436.03**		Explain in GSTR-9 Table 9 / 10–11 / 14 as applicable

Table 6 — ITC availed during the year

Table	Particulars	Taxable value	IGST	CGST	SGST / UTGST	Cess	Source / how worked out
6A	Total ITC availed through GSTR-3B (Table 4A)		434,487.79	519,756.97	519,756.97		GSTR-3B 4A(1)+(2)+(3)+(4)+(5) — 12 months)
6A1	ITC of the preceding year availed in this year (to 6A)		–	4,500.00	4,500.00		Suggested from the register: 2 invoice(s) of FY 2023-24 claimed in this year's 3B — must agree with last year's Table 12/13 (exc. Rule 37/37A reclaims → 6H)
6A2	Net ITC of this year (6A − 6A1)		**434,487.79**	**519,256.97**	**519,256.97**		Formula
6B	Inward supplies (net imports / RCM) — inputs		136,787.25	187,589.16	187,589.16		Pre-filled from the purchase register by ledger head — this year's invoices claimed in this year's 3B — confirm (see "3B ledger split")
6B	Inward supplies (net imports / RCM) — capital goods			220,320.00	220,320.00		Pre-filled from books — capital goods must be shown separately
6B	Inward supplies (net imports / RCM) — input services		221,760.54	94,677.73	94,677.73		Pre-filled from books
	Control: GSTR-3B 4A(5) "All other ITC" for the year		358,487.79	512,286.89	512,286.89		GSTR-3B 4A(5); 12 months)
	Difference: 6B (books split) + 6A1 − 3B 4A(5)		–	-6,000.00	-6,000.00		Should be nil — month-wise on the "Books vs 3B monthly" sheet
6C	Inward supplies from unregistered persons (RCM)		–	4,500.00	4,500.00		Suggested from books rows without GSTIN marked reverse charge — confirm
	Control: GSTR-3B 4A(3) ITC on reverse charge		–	11,430.08	11,430.08		GSTR-3B 4A(3), 12 months)
6D	Inward supplies from registered persons (RCM) + 6A(3) – 6C		–	**7,330.08**	**7,330.08**		Formula · books show RCM tax ₹25,432.34 on registered suppliers
6E	Import of goods (incl. SEZ) — split Inputs / Capital goods in the return		36,000.00	–	–		GSTR-3B 4A(1), 12 months)
6F	Import of services (excluding SEZ)		–	–	–		GSTR-3B 4A(2), 12 months)
6G	Input tax credit received from ISD		–	–	–		GSTR-3B 4A(4), 12 months)
6H	ITC reclaimed (Rule 37 / 37A)		–	–	–		GSTR-3B 4B(1) ITC reclaimed, 12 months) — keep only Rule 37/37A reclaims
6I	Sub-total (6B to 6H)		**434,487.79**	**514,256.97**	**514,256.97**		Formula
6J	**Difference (6I − 6A2) — should be nil**		–	5,000.00	-5,000.00		Formula
6K–6M	Transition credit / other ITC (TRAN-1, TRAN-2, ITC 01/02/02A)						Fill in if any
6O	Total ITC availed (6I + 6K to 6M)		**434,487.79**	**514,256.97**	**514,256.97**		Formula

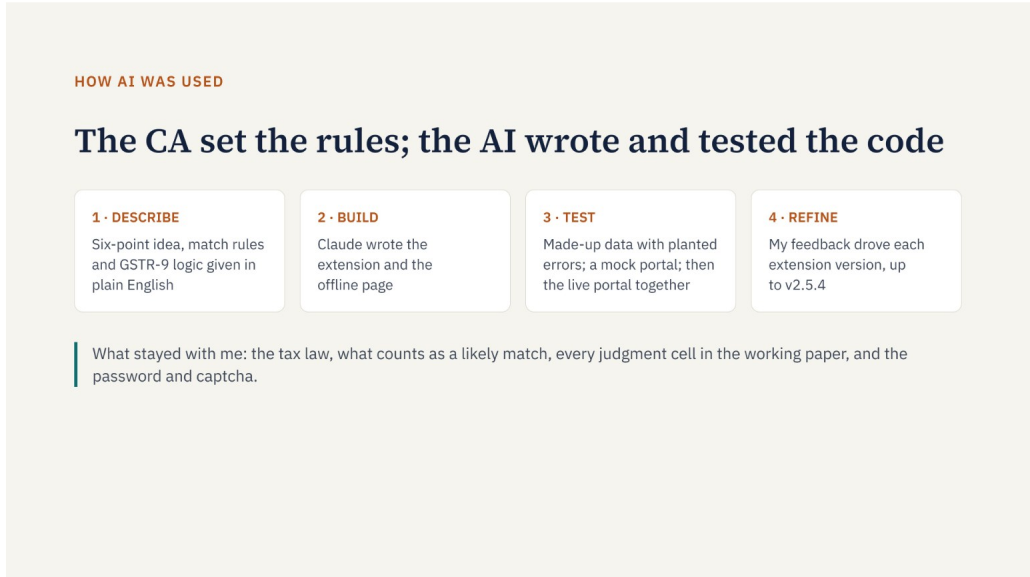
Table 7 — ITC reversed and ineligible

Table No.	Particulars	Taxable value	IGST	CGST	SGST / UTGST	Cess	Source / how worked out
7A	As per Rule 37 (payment not made within 180 days)						Trace check: 4 invoice(s), tax ₹39,073.62 unpaid after 180 days as on 30-11-2025 — see "RC risk flags"
7A1	As per Rule 37A (supplier has not paid tax / filed GSTR-3B)						Fill in — GSTR-2A "GSTR-3B filing status" shows suppliers who have not filed
7A2	As per Rule 38 (banks / financial institutions)						Fill in if applicable
7B	As per Rule 39 (QSD)						Fill in if applicable
7C	As per Rule 42 (common inputs / services)						Fill in if applicable
7D	As per Rule 43 (common capital goods)						Fill in if applicable

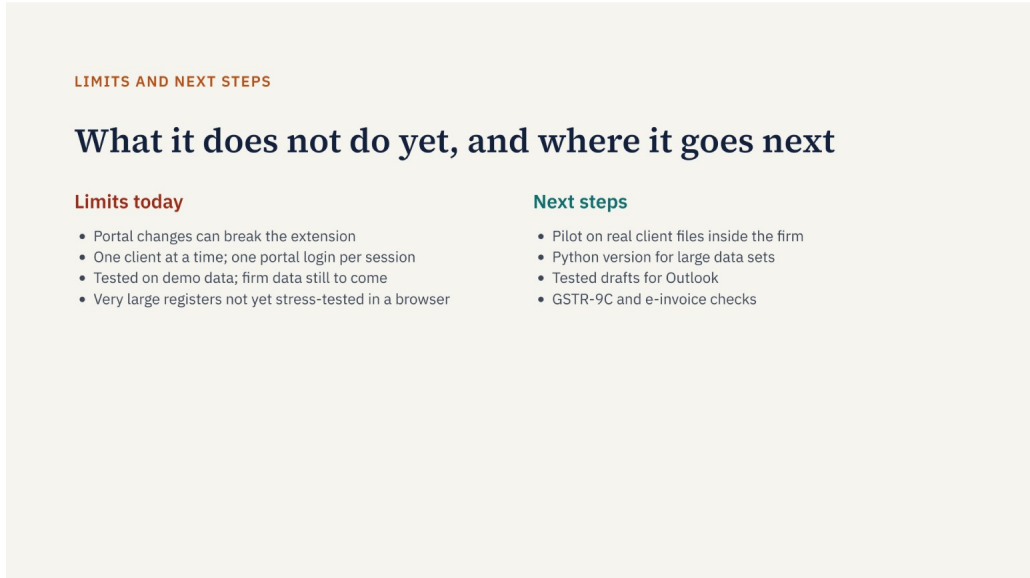
The output is an Excel working paper that a reviewer can trace. Yellow cells are left for the CA's judgment, and the totals are live formulas, so a change flows through. The 8D difference, 2B minus what was claimed, is explained invoice by invoice: blocked credit, credit claimed next year, invoices not in 2B, and so on, so the unexplained line should be nil. Risk flags point to section 17(5) items, section 16(4) time limits, Rule 37 payments beyond 180 days, reverse charge months, and months where 3B differs from the books.



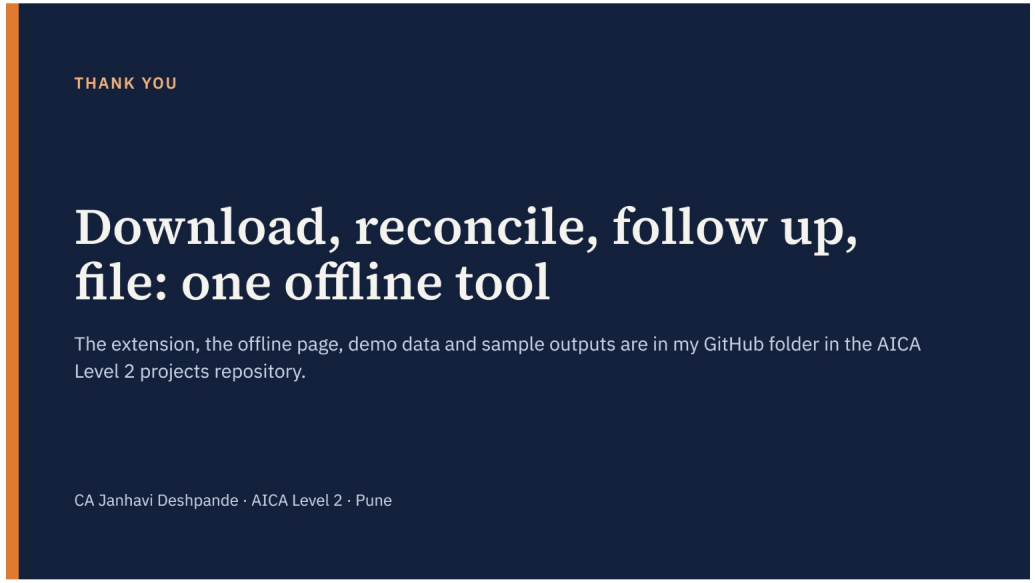
To test the tool I built demo data with known errors planted in it. On the quarterly demo the tool found 46 matched, 3 likely matched and 9 unmatched invoices, matching what was planted. On the full-year demo, the 8D difference was fully explained, with nothing left unexplained, next-year claims of about fourteen thousand rupees went to 8C, and eighteen ITC risk flags were raised for review. On the live portal the extension downloaded six months of GSTR-2B. And the offline page made no network request at all.



I built this with Claude, an AI assistant. I described the idea and the rules in plain English: what counts as a match, a likely match, how GSTR-9 tables are filled. Claude wrote the code, then tested it on made-up data with errors deliberately planted, and on a mock copy of the portal. We then tested the download together on the live portal, with me typing the password and captcha. Each problem I reported became a new version of the extension. The tax law and the judgment stayed with me; the AI did the programming.



There are honest limits. The extension follows the portal's screens, so a portal redesign can break it and it will need updating. It handles one client and one login at a time. It has been tested on demo data and on the live portal for downloads, but not yet on a full year of real client data, and very large purchase registers have not been stress-tested in the browser. Next, I will pilot it on real files inside the firm, build a Python version for large data, test the drafts in Outlook, and look at GSTR-9C.



To sum up: one tool that downloads the returns, reconciles them, matches the books with GSTR-2B, prepares the vendor follow-up and builds the GSTR-9 working paper, with client data never leaving the firm's computer. Everything I showed, the extension, the offline page, the demo data and the sample outputs, is in my GitHub folder. Thank you for watching.